LAURA RODRIGO HERNANDO

Full-Stack Developer · Angular · Python/FastAPI · Applied AI

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PROFILE

Full-stack developer with 3+ years of experience building data visualisation web applications with Angular, Python/FastAPI and PostgreSQL/PostGIS: interactive maps, charts, REST APIs and Docker and CI/CD deployments. I have worked on more than 8 applications, from the user interface to the database, and built an internal application end to end with Keycloak authentication. I bring generative AI into my work through data extraction with local vision models, RAG systems and coding agents.

TECHNICAL SKILLS

Frontend: Angular, TypeScript, HTML/CSS, ng-zorro, PrimeNG, Clarity, Leaflet, ECharts, Plotly.js, GSAP, That Open (IFC/BIM), i18n, Figma

Backend & data: Python, FastAPI, .NET, PostgreSQL + PostGIS (GeoJSON), Celery, data migrations (CSV/JSON), DBeaver, uv, Poetry

Security: Keycloak (setup, administration and 2FA), JWT, cross-application roles and permissions

DevOps & quality: Git, CI/CD, Docker, development and production environments, SonarQube, Cypress, pytest, Linux

Applied AI: Local LLMs (Ollama), vision models, RAG (LlamaIndex, LangGraph), LLM API integration (OpenAI, Anthropic), coding agents (Claude Code, Codex)

AI foundations: Machine learning and deep learning, neural networks, transformers and attention, embeddings; PyTorch, pandas, NumPy, Hugging Face, LangChain, Langfuse

Ways of working: Scrum, multidisciplinary teams, mentoring interns and juniors

PROFESSIONAL EXPERIENCE

Full-Stack Developer · CARTIF, Boecillo (Valladolid), Spain *Jun 2023 – present*

- Develop 8+ data visualisation and analysis web applications with Angular and Python/FastAPI, from the user interface to the database.
- Turn prediction models into web tools: parameter configuration, asynchronous execution (Celery) and results on maps (Leaflet) and charts (ECharts, Plotly.js).
- Design REST APIs and geospatial databases (PostgreSQL/PostGIS) and contribute to architecture decisions and frontend-backend contracts.
- Ensure quality and continuous delivery with E2E (Cypress) and backend (pytest) tests, SonarQube analysis, Docker containers and CI/CD pipelines.
- Work in Scrum within multidisciplinary teams, and train and mentor interns and junior developers.

SELECTED PROJECTS

Internal tools catalogue (in progress) · Full-stack · technical owner

Designed and built the whole application: PostgreSQL/PostGIS data model, FastAPI REST API and Angular frontend. Set up Keycloak with two-factor authentication and embedded, with shared permissions, another CARTIF application I built with a colleague and adapted for the integration.

Receipt and invoice extraction with AI · Backend & AI · team of 2

FastAPI API that classifies photos of receipts and invoices with a local vision model (Ollama) and extracts category, amount and currency as structured JSON, rejecting unreadable or invalid documents. Data never leaves the company's infrastructure.

Climres (in progress) · Backend

Geospatial PostgreSQL/PostGIS database for GeoJSON data, FastAPI REST API and Python migration scripts to integrate CSV and JSON data.

Smarteestory (in progress) · Frontend

Building digital twin: IFC/BIM model viewer (That Open), sensor data from an external API and a full document manager.

Spandam and RethinkAction · Frontend

Decision-support tools with interactive maps, charts and asynchronous calculations. On Spandam I proposed design improvements on top of Figma and built the bilingual interface and the Cypress E2E tests.

Other frontend projects · Hydro4u and Rehouse (in progress), Build Space, Nevermore

Interfaces with Angular, Leaflet, ECharts and component libraries (ng-zorro, PrimeNG, Clarity): prediction model visualisation, document management, collaborative projects and mapping.

Web Developer Intern (training contract) · CARTIF, Boecillo (Valladolid), Spain *Mar 2023 – May 2023*

- Built end-to-end features for the corporate intranet with Angular, .NET and PostgreSQL, working with secure JWT-based authorisation and CI/CD deployment.

PERSONAL PROJECTS

Interactive portfolio with a RAG assistant

Personal website with a 3D galaxy (Next.js, React Three Fiber) and an assistant that answers questions about my profile: RAG with LangGraph and hybrid search, compatible with local (Ollama) and cloud models (OpenAI, Anthropic).

Educational RAG chatbot (in progress)

Assistant for an educational project built on a custom LlamaIndex RAG system, designed to run on both local and cloud models.

EDUCATION & TRAINING

Fundamentals and tools for generative AI (in-company training) · Universidad Politécnica de Madrid *2026*

Transformers and attention, model training (overfitting, underfitting, data augmentation), Hugging Face, LangChain, LangGraph, Langfuse and RAG.

Machine Learning and Deep Learning (in-company training) · Universidad de Valladolid *2025*

Mathematical foundations and algorithms of ML and DL: neural networks, backpropagation, activation functions, embeddings and transformers. Hands-on work with PyTorch, pandas and NumPy.

Learn FastAPI from scratch · Udemy *2025*

Higher Vocational Diploma in Web Application Development · IES Galileo, Valladolid *2021 – 2023*

LANGUAGES

Spanish: native · English: professional working proficiency

AVAILABILITY

Remote, asynchronous work, either employed or as a freelancer on a fixed commitment (a set number of hours per week for an agreed period).